



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Off Road Vehicles		
Course Code:	PE-AUE801A	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand the various off-road vehicle and their systems and features.		
2	To understand the hydraulic and pneumatic circuits for various off-road vehicles		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Shovels: Types of Shovel, Construction details of diesel, electric and hydraulic shovel. Operating principles and operating cycle. Production capacity and cost of production.	6
2	Draglines: Types of Dragline, Construction and Operating cycle, Production capacity and cost of production.	6
3	Dumpers: Types of Dumpers. Construction and operating cycle, Carrying capacity, Matching with Shovel capacity and cost of production.	5
4	Dozer and Grader: Different types of Dozer, Construction and operation, dozer capacity, grader and its construction, application of dozer and grader.	6
5	Tractor and Tractor Units: Tractors in Earth Moving, Application of Tractors, Rating of Tractors, Wheel and Crawler Tractor, Recent Trend in Tractor Design, Power shift Transmission and Final drive in Caterpillar Tractor, Control Mechanism of a Caterpillar.	7
6	Fork Lift Truck and Road Roller: Types, Layout and lifting mechanism of Fork Lift Truck, Construction and working Principle of Fork Lift truck, Types, Layout, operation & maintenance of Road Roller.	6
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Understand the types, special features, design methodology, working principle, application of various off-road vehicles.
2	Identify various systems & subsystems of earth moving machines & agricultural machines.
3	Understand the hydraulic and pneumatic circuits for various off-road vehicles
4	Understand preventive maintenance of earth moving machines & agricultural machines

Learning Resources:	
1	Schmitt, R., Schexnayder, C. J, Aaron Cohen, A, Nichols, H, Day, D “Moving the Earth: Excavation Equipment, Methods, Safety, and Cost”. McGraw Hill Publications, Seventh Edition”, 2019
2	Marek Sokolski, “Mining Machines and Earth-Moving Equipment Problems of Design, Research and Maintenance”. Springer Book, 2020
3	Abrosimov K., Branberg A. and Katayer K., Road Making Machinery, MIR Publishers, Moscow, 1971
4	De, A., Latest Development of Heavy Earth Moving Machinery, Annapurna Publishers, Dhanbad, 1995.
5	Nichols H.L.Jr., Moving the Earth, Galgotia Publishing House, New Delhi, 1962.
6	Rudnev V.K., Digging of Soils by Earthmover with Power Parts, Oxanian Press Pvt. Ltd., New Delhi, 1985.



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Course Name:	Automotive Air Conditioning		
Course Code:	PE-AUE801B	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Thermal Engineering
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To familiarize with the terminology associated automotive air conditioning systems.		
2	To acquire the skills required to model, analyses and design of air conditioning processes and components.		
3	To understand the basics of psychrometry and practice of applied psychometrics.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Basic air conditioning fundamentals: Air conditioning principles, Air-conditioning types, temperature and pressure fundamentals, Human comfort.	3
2	Air-conditioning System: Simple vapour compression refrigeration system (V.C.R.S), Driers, Lubricants, Air-conditioning components and controls: compressors, condenser, evaporators, valves, electrical circuits and devices, Refrigerants and their properties.	8
3	Psychometrics: Psychometric properties and processes, sensible and latent heat loads, characterization and SHF load for ventilation and filtration, concepts of SHF and ESHF and ADP, concepts of human comfort and effective temperature.	6
4	Distribution duct system, sizing, supply I return ducts, type of grills, diffusers, ventilation, air noise level, layout of duct systems for automobiles and their impact on load calculations.	6
5	Load Analysis: Heat transfer from exterior wall, passenger, Equipment and infiltrated air. Heater system for winter conditioning, Requirement of air and air distribution systems, duct design, duct systems.	8
6	Air conditioner maintenance & service: causes of air conditioner failure, leak testing guide, discharging the system, Evacuating the system, charging the system, servicing heater system, removing & replacing components, trouble shooting of air conditioning system, compressor service.	5
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Identify various components of Vehicle Air conditioning and heating system.
2	Operate manually and automatic Air conditioning and heating system.
3	Apply various concepts related to Air conditioning and heating system.
4	Diagnose various faults in air conditioning system by using suitable tools and instruments.
5	Follow safety rules while servicing of Air conditioning and heating system.

Learning Resources:	
1	Stoecker W.F. and Jones J.W., Refrigeration & Air-conditioning, McGraw Hill Publishing Company Limited, 1982
2	Lung P., Automotive Air Conditioning, C.B.S. Publisher & Distributor, New Delhi.
3	Giri N.K., Automotive Technology, Khanna Publishers, 2004.
4	William H. Crouse & Donald L. Anglin, "Automotive Air Conditioning. Mc Graw Hill, Inc., 1990.
5	Paul Lung, "Automotive Air Conditioning, C.B, S. Publisher & Distributor, Delhi.
6	Paul Weisler, "Automotive Air Conditioning, Reston PublishingCo.Inc.1990.



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Course Name:	Battery Technology and Management Systems		
Course Code:	PE-AUE801C	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To explain fundamental knowledge on electrochemical energy storage systems considering the operation and design of various battery technologies.		
2	To find the requirement of batteries for automotive application combined with environment policy considerations.		
3	To define learner to batteries, its parameters modeling and charging requirements.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to Battery Terminologies, Electrochemistry, Cells & Batteries, Nominal voltage and capacity, C rate, Energy and power, Cells connected in series, Cells connected in parallel, Electrochemical and lithium-ion cells, Rechargeable cell, Charging and Discharging Process, Overcharge and Undercharge, Modes of Charging, Charging and Discharging for EV Application, Fuel cell,	7
2	Battery Management System Requirement: Introduction and BMS functionality, Battery pack topology, BMS Functionality, Voltage Sensing, Temperature Sensing, Current Sensing, BMS Functionality, High-voltage contactor control, Isolation sensing, Thermal control, Protection, Communication Interface, Range estimation, State-of charge estimation, Cell total energy and cell total power, Thermal Management, Types of temperature sensors, Thermal model of battery pack, Cooling materials and methods.	8
3	Battery State of Charge and State of Health Estimation, Cell Balancing: Battery state of charge estimation (SOC), voltage-based methods to estimate SOC, Model-based state estimation, Battery State of Health (SOH) Estimation, Lithium-ion aging: Negative electrode, Lithium-ion aging: Positive electrode, Cell Balancing, Causes of imbalance, Circuits for balancing	7
4	Modelling and Simulation: Mathematical Modeling, Equivalent-circuit models (ECMs), Physics-based models (PBMs), Empirical modeling approach, Physics-based modeling approach, Simulating an electric vehicle, Vehicle range calculations, Simulating constant power and voltage, Simulating battery packs,	7
5	Design of battery BMS: Design principles of battery BMS, Effect of distance, load, and force on battery life and BMS, energy balancing with multi-battery system	7
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Interpret the role of battery management system.
2	Identify the requirements of Battery Management System.
3	Understand the requirements of battery systems for automotive applications and understand the modelling of battery systems.
4	Interpret the concept associated with battery charging / discharging process.

Learning Resources:	
1	Bard, Allen J., and Larry R. Faulkner. Electrochemical Methods: Fundamentals and Applications. 2nd ed., Wiley– VCH, Verlag, GmbH, 2000.
2	Masataka Wakihara and Osamu Yamamoto, Lithium ion Batteries Fundamental and Performance, Wiley–VCH, Verlag GmbH, 1999.
3	Robert A. Huggins, Advanced Batteries – Materials science aspects, Springer, 2009.
4	Davide Andrea, Battery Management Systems for Large Lithium-ion Battery Packs, Artech House, 2010.
5	Bergveld, H.J., Kruijt, W.S., Notten, P.H.L, Battery Management Systems -Design by Modelling, Philips Research Book Series, 2002.



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Course Name:	Fuel cell Technology		
Course Code:	PE-AUE801D	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Thermodynamics & Heat Transfer
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart knowledge on concepts and application of fuel cell technology		
2	To equip the students to analyse various components of Fuel cells.		
3	To understand the working of Standalone Fuel cells and hydrogen storage devices		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to Fuel Cells: Introduction – working and types of fuel cell – low, medium and high temperature fuel cell, liquid and methanol types, proton exchange membrane fuel cell solid oxide, hydrogen fuel cells– thermodynamics and electrochemical kinetics of fuel cells.	7
2	Fuel Cells for Automotive Applications: Fuel cells for automotive applications – technology advances in fuel cell vehicle systems –onboard hydrogen storage – liquid hydrogen and compressed hydrogen – metal hydrides, fuelcell control system – alkaline fuel cell – road map to market.	6
3	Fuel Cell Components and Their Impact on Performance: Thermodynamic aspects of Electrochemical Energy conversion- Cell efficiency – Factors affecting the efficiency of Electrochemical Energy conversion–Fuel cell performance characteristics – current/voltage, voltage efficiency and power density, ohmic resistance, kinetic performance, mass transfer effects – membrane electrode assemblycomponents, fuel cell stack, bi-polar plate, humidifiers and cooling plates.	10
4	Fueling: Hydrogen storage technology – pressure cylinders, liquid hydrogen, metal hydrides, carbonfibers – reformer technology – steam reforming, partial oxidation, auto thermal reforming – COremoval, fuel cell technology based on removal like bio-mass.	6
5	Fuel Cycle Analysis: Introduction to fuel cycle analysis – application to fuel cell and other competing technologieslike battery powered vehicles, SI engine fueled by natural gas and hydrogen and hybrid electricvehicle.	7
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Describe the working principles of Fuel cells and its component
2	Develop clear understanding about functioning and types of Fuel cells
3	Familiarize with the reaction kinetics involved in fuel cell
4	Build skill on thermodynamics concepts involved in fuel cell.

Learning Resources:	
1	Fuel Cells for automotive applications – professional engineering publishing UK, 2004.
2	Ryan O’Hayre, Suk Won Cha and Whitney Colella, Fuel cell fundamentals, Second edition, John Wiley and Sons, 2016
3	Viswanathan, B and M Aulice Scibioh, Fuel Cells – Principles and Applications, Universities Press, 2006
4	Supramaniam Srinivasan, Fuel cells: From fundamental to application, First Edition, Springer,2010
5	Bent Sorensen, Hydrogen and Fuel cells, Academic Press, 2011
6	Noriko Hikosaka Behling, Fuel cells, Elsevier Publishers, 2012
7	Fuel Cell Technology Handbook, Gregor Hoogers, SAE International CRC Press, 2003



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Course Name:	Thermal Management of Hybrid Systems		
Course Code:	PE-AUE802A	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Thermodynamics & Heat Transfer
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To study the basic knowledge of thermal management system of different component of hybrid systems.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Brief Review of Thermodynamics, Fluid Mechanics, and Heat Transfer: First Law of Thermodynamics for open and closed systems; internal energy, enthalpy, and specific heat - Second Law of Thermodynamics for closed systems; Tds equations, Gibbs function - Fluid mechanics: laminar vs. turbulent flow, internal flow relationships, Navier Stokes equations - Heat transfer: simple conduction, convection, and radiation relationships; Nusselt number relationships for convective heat transfer; energy equation.	9
2	Thermal Management of Motors: Motor Sizing vs Heat Generation - Operational Temperature Limitations of Electrical Insulation - Design concepts for Heat Extraction in Motors for HEV/EV systems - Modelling and simulation of heat transfer in motors - Rendering of Heat extraction solutions - Sensors and Protection solutions.	7
3	Thermal Management for Batteries and Power Electronics: Introduction - Thermal control in vehicular battery systems: battery performance degradation at low and high temperatures - Passive, active, liquid, air thermal control system configurations for HEV and EV applications - Battery Heat Transfer - Introduction to battery modeling: tracking current demand, voltage, and State of Charge as functions of time for given drive cycles - Development of thermodynamic relationships for cell heat generation - Lumped cell and pack models for transient temperature response to drive cycles - Model parametric study results	10
4	Thermal Management Systems: Overall energy balance to determine required flowrates - Determination of convection and friction coefficients for air and liquid systems in various geometric configurations: flow around cylinders, flow between plates, flow through channels - Development of a complete thermal system model and parametric study results - Temperature control and heat transfer using phase change materials - Thermal Management of Power Electronics.	10
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Apply the concepts of thermodynamics to understand the thermal management system
2	Understood and apply the thermal management system of different component of hybrid systems like Motors, Batteries and Power Electronics
3	Develop the thermal system model due to flow of heat & phase change over different types of geometrical shapes and different materials.

Learning Resources:	
1	T. Yomi Obidi, “Thermal Management in Automotive applications”, 2015, SAE International.
2	Jerry Sergeant, Al Krum, “Thermal Management Handbook: For Electronic Assemblies Hardcover”, 1998, Mc Graw- Hill
3	Younes Shabany, “Heat Transfer: Thermal Management of Electronics Hardcover” 2010, CRC Press.
4	“Vehicle thermal Management Systems Conference Proceedings”, 1st Edition; 2013, Coventry Techno centre, UK
5	Nag. P. K, “Engineering Thermodynamics”, 6th Edition, Tata McGraw Hill Education, New Delhi, 2017.



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Course Name:	Noise, Vibrations and Harshness		
Course Code:	PE-AUE802B	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To study basic concepts of noise, vibration and harshness and their effects.		
2	To study various methods of Vibration control.		
3	To study and analyze sounds and detection of noise from automobiles.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Basics of Vibrations: Basic Concepts, Mathematical Models, System characteristics and response, Single and Multi DOF systems.	6
2	Vibration control: Isolators, Tuned absorbers, Untuned viscous dampers, Applications: single cylinder engines, multi cylinder engine, Simple rubber engine mounts, Hydro elastic mounts, Semi active mounts and active mounts, Mass elastic models and measurements, Limits for passenger comforts.	8
3	Sound & sound measurement: Fundamentals of acoustics, General sound propagation, Plane wave propagation, Spherical wave propagation, Human response to sound– the audible range, Sound measurement, Instrumentation, Sound level meters, Frequency intensity analyzers, Real time measurements.	8
4	Automotive noise: Automotive noise criteria, Drive by noise test, Noise from stationary vehicles, Interior noise in vehicles, Automotive noise, Sources and control methods: Engine noise, Transmission noise, Intake and exhaust noise, Aerodynamic noise, Tyre noise, Brake noise.	6
5	General noise control principles: Sound in enclosures, Sound energy absorption, Sound transmission through barriers.	4
6	Harshness: Causes, Frequency limits.	4
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Identify and analyze vibrations and noise coming out of automobiles.	
2	Investigate level of harm caused by noise and harshness and to provide measures to control it.	

Learning Resources:	
1	Rao S.S., Mechanical Vibrations, Addison Wesley Longman, New Delhi, 1995.
2	Noise Vibration & Harshness And Safety By K. V. Fadadu (Author), B. H. Kadiya (Author) Paperback – 1 January 2017
3	SAE, Automobiles and Pollution, SAE Transaction, 1995.
4	Seto, Mechanical Vibrations, Schaum Outline Series, McGraw Hill Book Company, New York, 1990.
5	Noise Vibration and Harshness Paperback – 1 January 2011 by Abdul Samad & Ranjeet Singh Rathore" Shyam Sunder Suthar, Y. B. Mathur (Author)
6	Thomson W.T., Theory of Vibration with Applications, CBS Publishers and Distributors, New Delhi, 1990.
7	Mallik A.K., Principles of Vibration Control, Affiliated East-West Press (P) Ltd., New Delhi, 1990.
8	Grover G.K., Mechanical Vibrations, New Chand and Brothers, Roorkey, 1989.
9	Morse T. and Hinkle, Mechanical Vibration, Prentice Hall of India Ltd., New Delhi, 1987.



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Course Name:	Electric Motors for Automotive Applications		
Course Code:	PE-AUE 802C	Category:	Professional Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic electrical engineering, control system
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand the basics of electric motors		
2	To understand the performance of electric motors required in automotive industry		
3	To understand the drive modules of different electric motors.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Motors Characteristics and DC motor: Requirement of EV motors, Comparison of EV motors, Basics of DC Motor, Torque speed characteristics, DC Motor dynamics, speed control methods, starting and braking, Four quadrant operation, Early Ward Leonard control system - Main features - generator - merits - reverse motion - modified Ward Leonard control system -Main features - modifications. Modern electric drives -Main features -performance characteristics - advantages and limitations. DC-DC Converter, Two quadrant DC Chopper, converter-fed dc motor drive. Principle of operation of BLDC motor – Types of constructions – Surface mounted and interior type permanent magnet DC Motors – Torque and EMF equations for Square wave & Sine wave for PMBLDC Motor – Torque – Speed characteristics of Square wave & Sine wave for PMBLDC Motor – Merits & demerits of Square wave & Sine wave for PMBLDC Motor – Performance and efficiency – Applications.	12
2	DC Motor Dynamics & Control Current Loop Control, Speed Control Loop Dynamical System Control: Gain & Phase Margins, PD Controller, PI Controller, Selecting PI Gain for Speed Controller, PI Controller Design, PI Controller with Reference model, Comparison of conventional PI Controller with PI controller with Reference Model, 2 DOF Controller with Internal Model Control, Load Torque Observer, Feedback Linearization, Simplified Modeling of Practical Current Loop.	8

3	Induction Motor and speed control Rotating Magnetic Field, Basics of Induction motor, Speed-Torque Curve Leakage inductance, circle diagram, current displacement (double cage rotor), line starting, Dynamic modelling of Induction motor, Rotor Field oriented control, Stator Field Oriented Control, Field Weakening Control, Variable Voltage Variable Frequency Control. Three Phase Inverter Based Induction Motor Drive, Equal Area PWM, Three Phase Auxiliary resonant snubber (ARS) Inverter Type (ZVC & ZCS), Single Phase ARS Inverter Topology, Speed Control of Induction Motor.	7
4	Electric Propulsion Systems: DC motor drives, induction motor drives, permanent magnet motor drives and switched reluctance motor drives.	9
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Understand requirement of electric motors.	
2	Understand the operation, performance of electric motors	
3	Understand suitability of electric motor & their control	

Learning Resources:		
1	Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay and Ali Emadi, “Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design”, CRC Press, 2009.	
2	K Wang Hee Nam: AC Motor Control & Electrical Vehicle Application, CR Press, Taylor & Francis Group, 2019	
3	C.C Chan, K.T Chau: Modern Electric Vehicle Technology, Oxford University Press Inc., New York 2001	



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Course Name:	Tribology		
Course Code:	OE-ME801F	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Fluid Mechanics.
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To provide students with the fundamental knowledge in the field of Industrial tribology.		
2	To provide basic concepts in the design of automotive lubrication system.		
3	To provide knowledge of friction and wear mechanism in automotive system.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to Tribology: Introduction to Tribology, Tribology in design, Tribology in industry, economic aspects of Tribology, lubrication, basic modes of lubrication, lubricants, properties of lubricants-physical and chemical, types of additives, extreme pressure lubricants, recycling of used oils and oil conservation, disposal of scrap oil, oil emulsion. Types of sliding contact bearings, comparison of sliding and rolling contact bearings.	7
2	Friction and Wear: Friction: Introduction, laws of friction, kinds of friction, causes of friction, friction measurement, theories of friction, effect of surface preparation. Wear: Types of wear, various factors affecting wear, measurement of wear, wear between solids and liquids, theories of wear.	5
3	Hydrodynamic lubrication: Theory of hydrodynamic lubrication, mechanism of pressure development in oil film, two-dimensional Reynold's equation, infinitely long journal bearing, infinitely short journal bearing, finite bearing. Hydrodynamic thrust bearing: Introduction, flat plate thrust bearing, pressure equation, load, centre of pressure, friction in tilting pad thrust bearing.	7
4	Hydrostatic Lubrication: Hydrostatic lubrication: Basic concept, advantages and limitations, viscous flow through rectangular slot, load carrying capacity and flow requirement of hydrostatic step bearing, energy losses, optimum design of step bearing. Compensators and their actions. Squeeze film lubrication: Introduction, circular and rectangular plates approaching a plane	6
5	Elasto-hydrodynamic Lubrication and Gas Lubrication: Electrohydrodynamic Lubrication: Principle and application, pressure-viscosity term in Reynolds equation, Hertz theory. Ertel Grubin Equation. Gas lubrication: Introduction, merits and demerits, applications. Lubrication in metal working: Rolling, forging, drawing and extrusion.	6

	Bearing materials, bearing constructions, oil seals, shields and gaskets	
6	Surface Engineering: Introduction to surface engineering, concept and scope of surface engineering, manufacturing of surface layers, solid surface geometrical, mechanical and physic chemical concepts, superficial -layer, development of concept, structure of superficial layer, general characteristics of superficial layer, obtained by machining, strengthening and weakening of superficial layer.	5
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Apply knowledge of tribology for industrial component design	
2	Apply design concepts practically for automotive lubrication systems.	

Learning Resources:	
1	Cameron A., Basic Lubrication Theory, Wiley Eastern Ltd. Publish 1987.
2	Wen S., Principles of Tribology, Wiley. June 2017.
3	Majumdar B.C., Introduction to Tribology and Bearings, S. Chand and Company Ltd., New Delhi. December 2010.
4	Halling J., Principles of Tribology, McMillan Press Ltd. December 1978
5	Bhushan B. and Gupta B.K., Handbook of Tribology: Materials, Coatings and Surface Treatments, McGraw-Hill, 1991.
6	Davis J., Surface Engineering for Corrosion and Wear Resistance, Woodhead Publishing, 2001.



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Course Name:	Internet of Things		
Course Code:	OE-IT801C	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart necessary and practical knowledge of components of the Internet of Things and develop skills required to build real-life IoT-based projects.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to IoT Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals, Devices and gateways, Data management, Business processes in IoT, Everything as a Service (XaaS), Role of Cloud in IoT, Security aspects in IoT.	10
2	Elements of IoT Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for Communication Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.	10
3	IoT Application Development Solution framework for IoT applications- Implementation of Device integration, Data acquisition and integration, Device data storage- Unstructured data storage on cloud/local server, Authentication, authorization of devices.	10
4	IoT Case Studies IoT case studies and mini projects based on Industrial automation, Transportation, Agriculture, Healthcare, Home Automation.	6
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Understand the internet of Things and its hardware and software components.	
2	Interface I/O devices, sensors & communication Modules.	
3	Remotely monitor data and control devices.	
4	Develop real-life IoT based projects.	

Learning Resources:	
1	Jeeva Jose, “Internet of Things”, 1 st edition, Khanna Publishing House, 2018
2	Vijay Madiseti, Arshdeep Bahga, “ Internet of Things :A Hands on Approach”, 1 st edition, Orient Blackswan Private Limited, 2015
3	Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, “Introduction to Internet of Things: A practical Approach”, ETI Labs
4	Pethuru Raj and Anupama C. Raman, “The Internet of Things: Enabling Technologies, Platforms, and Use Cases”, CRC Press
5	Adrian McEwen, “Designing the Internet of Things”, Wiley
6	Raj Kamal, “Internet of Things: Architecture and Design”, McGraw Hill
7	Cuno Pfister, “Getting Started with the Internet of Things”, O Reilly Media



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Course Name:	Soft Computing		
Course Code:	OE-CS801H	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics III(BS-M303)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To give students knowledge of soft computing theories fundamentals.		
2	To enable students to learn and apply artificial neural networks, fuzzy systems, and genetic algorithms for solving real-life problems		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Neural Networks– I: (Introduction and Architecture) Neuron, Nerve Structure and Synapse, Artificial Neuron and its Model, Activation Functions, Neural Network Architecture: Single Layer and Multilayer Feed Forward Networks, Recurrent Networks. Various Learning Techniques; Perception and Convergence Rule, Auto Associative and Hetro-Associative Memory.	7
2	Neural Networks– II: (Back Propagation Networks) Architecture: Perceptron Model, Solution, Single Layer Artificial Neural Network, Multilayer Perception Model; Back Propagation Learning Methods, Effect of Learning Rule Co-Efficient; Back Propagation Algorithm, Factors Affecting Back Propagation Training, Applications.	8
3	Fuzzy Logic– I: (Introduction) Basic Concepts of Fuzzy Logic, Fuzzy Sets, and Crisp Sets, Fuzzy Set Theory and Operations, Properties of Fuzzy Sets, Fuzzy and Crisp Relations, Fuzzy to Crisp Conversion.	7
4	Fuzzy Logic– II: (Fuzzy Membership, Rules) Membership Functions, Inference in Fuzzy Logic, Fuzzy If-Then Rules, Fuzzy Implications, and Fuzzy Algorithms, Fuzzification and Defuzzification, Fuzzy Controller, Industrial Applications.	7
5	Genetic Algorithm (GA): Basic Concepts, Working Principle, Procedures of GA, Flow Chart of GA, Genetic Representations, (Encoding) Initialization and Selection, Genetic Operators, Mutation, Generational Cycle, Applications.	7
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Explain various neural network architectures, backpropagation neural networks, and their application
2	Explain the concepts of fuzzy systems, fuzzy membership, and rules
3	Illustrate the genetic algorithms and their applications.
4	Apply concepts of neural networks, fuzzy logic, and genetic algorithm for solving problems in real life

Learning Resources:	
1	Rajasekaran S. and Pai V.G.A., Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications, Prentice Hall of India, 2003.
2	S N Sivanandam, S. Sumathi, “Principles of Soft Computing”, 3 rd edition, John Wiley & Sons,2018
3	S. Roy & U. Chakraborty, “Soft Computing: Neuro Fuzzy and Genetic Algorithms”, 1 st Edition, Pearson,2013
4	D.K. Pratihari “Soft Computing: Fundamentals and Applications”, Narosa Publishing House Pvt. Ltd. - New Delhi,2015
5	S. Kumar, “Neural Networks: A Classroom Approach”,2 nd Edition, TMH, 2017
6	S. Haykin, “Neural Networks: A Comprehensive Foundation”2 nd Edition, Prentice Hall.
7	G.J. Klir and B. Yuan, “Fuzzy Sets and Fuzzy Logic: Theory and Applications”, 1 st Edition, Prentice Hall,2015
8	D.E. Goldberg, “Genetic Algorithms in search, Optimization & Machine Learning”, Pearson ,2008



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Course Name:	Computational Fluid Dynamics		
Course Code:	OE-ME801B	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Engineering Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand mathematical characteristics of partial differential equations.		
2	To understand basic properties of computational methods– accuracy, stability, consistency.		
3	To learn computational solution techniques for time integration of ordinary differential equations.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to CFD: Evaluation Method, References and Application of CFD in Engineering.	3
2	Governing Equations and Boundary Conditions: Introduction to models of flow, conservation law of physics, derivation of Continuity, Momentum and Energy equation in Cartesian coordinate system, Transformation of these equations from Non conservative form to conservative, Implementation of boundary condition – Inlet, Outlet and Wall boundary condition.	10
3	Numeric Discretization Process: Concept and structure, Methods of deriving the discretized equations, Finite difference, Finite Volume Methods, stability criteria, errors in calculation, Mathematical behaviour of PDE's, structured grid, staggered grid, Mesh less Techniques.	9
4	Solution Algorithm: One way and Two-way coordinates, Explicit Taylor series expansion, The four basic rules for the control volume formulation, staggered grid Pressure correction as Poisson's pressure equation, Quick, Simple and PISO Algorithms, TMDDA, Point Iterative Methods, Explicit Methods– Crank Nicolson, Implicit Methods. Usage of Plotting software (open source, commercial), Contour plots, velocity vectors and Heat maps	14
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Develop mathematical characteristics of partial differential equations.
2	Choose basic properties of computational methods– accuracy, stability, consistency.
3	Analyze computational solution techniques for time integration of ordinary differential equations.
4	Build computational solution techniques for various types of partial differential equations.
5	Identify how to computationally solve Euler and Navier-Stokes equations
6	Build programming and graphic skills to conduct the flow field calculations and data analysis

Learning Resources:	
1	Hirsch C., Numerical Computation of Internal and External Flows, 2nd Ed., ButterworthHeinemann, 2007, ISBN 9780750665940 (E-Book available).
2	Pletcher R.H., Tannehill J.C. and Anderson D., Computational Fluid Mechanics and Heat Transfer, 3rd Ed., CRC Press, 2011.
3	Moin P., Fundamentals of Engineering Numerical Analysis, 2nd Ed., Cambridge University Press, 2010.
4	Introduction to Computational Fluid Dynamics: Development, Application and Analysis (Ane/Athena Books) Hardcover – 25 November 2016 by Atul Sharma.
5	Ferziger J.H. and Peric, M., Computational Methods for Fluid Dynamics, 3rd Ed., Springer, 2002.
6	LeVeque R., Numerical Methods for Conservation Laws, Lectures in Mathematics, ETH Zurich, Birkhauser, 1992.



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Course Name:	Entrepreneurship Development		
Course Code:	OE-HU801D	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	Acquiring Entrepreneurial spirit and resourcefulness.
2	Familiarization with various uses of human resource for earning dignified means of living.
3	Understanding the concept and process of entrepreneurship - its contribution and role in the growth and development of individual and the nation.
4	Acquiring entrepreneurial quality, competency, and motivation.
5	Learning the process and skills of creation and management of entrepreneurial venture.

Course Contents:

Mod ule No.	Description of Topic	Contact Hrs.
1	Introduction to Entrepreneurship and Start – Ups: Definitions, Traits of an entrepreneur, Intrapreneurship, Motivation; Types of Business Structures, Similarities/differences between entrepreneurs and managers.	7
2	Business Ideas and their implementation: Discovering ideas and visualizing the business; Activity map; Business Plan	6
3	Idea to Start-up: Market Analysis – Identifying the target market, Competition evaluation and Strategy Development, Marketing and accounting, Risk analysis	6
4	Management: Company's Organization Structure, Recruitment and management of talent. Financial organization and management	6
5	Financing and Protection of Ideas: Financing methods available for start-ups in India; Communication of Ideas to potential investors – Investor Pitch; Patenting and Licenses	6
6	Exit strategies: Exit strategies for entrepreneurs, bankruptcy, and succession and harvesting strategy.	5
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Understanding the dynamic role of entrepreneurship and small businesses
2	Organizing and Managing a Small Business & Financial Planning and Control
3	Demonstrate knowledge of the Forms of Ownership for Small Business with strategic Marketing Planning

Learning Resources:	
1	The Startup Owner's Manual: The Step-by-Step Guide for Building a Great Company, Steve Blank and Bob Dorf, K & S Ranch
2	The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Eric Ries, Penguin UK
3	Demand: Creating What People Love Before They Know They Want It, Adrian J. Slywotzky with Karl Weber, Headline Book Publishing
4	The Innovator's Dilemma: The Revolutionary Book That Will Change the Way You Do Business, Clayton M. Christensen, Harvard business



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Course Name:	Robotics and Robot Application		
Course Code:	OE-ME801E	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart knowledge about the engineering aspects of Robots and their application.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Basic concepts- Robot anatomy- Manipulators- Precision movement, robot specifications and Work volume, Types of Robot drives- Basic robot motions- Point to point control, continuous path contour.	6
2	Robot Kinematics and Dynamics: Forward and inverse kinematics, Translation and Rotation Representation, Coordinate transformation, DH parameters, Jacobian, Singularity, and Statics; Dynamic Modelling: Equations of motion: Euler-Lagrange formulation	8
3	End Effectors: End effectors- classification- mechanical, magnetic, vacuum and adhesive gripper- gripper force analysis and design. Robot control- Unit control system concept- servo and non-servo control of robot joints, adaptive and optimal control.	4
4	Sensors: Sensor devices, Types of sensors-contact, position and displacement sensors, Force and torque sensors- Proximity and range sensors- acoustic sensors- Robot vision systems- Sensing and digitizing- Image processing and analysis.	6
5	Robot Programming: Robot language classification- programming methods-off and online programming- Lead through method-Teach pendent method-VAL systems and language, simple program.	6
6	Industrial Application: Application of robots- Material handling- Machine loading and unloading, Assembly, Inspection, Welding, spray painting, Mobile robot, Microbots- Recent developments in robotics- safety consideration.	6
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Get familiarization with the basics of robots control system	
2	Get familiarization with end effectors, sensor technology and industrial application of robot	

Learning Resources:	
1	Deb, S.R., Deb Sankha, Robotics Technology and Flexible Automation, McGraw Hill Publishing Company Limited, New Delhi, 2017
2	Nicholas Odrey, Mitchell Weiss, Mikell Groover, Roger Nagel, Ashish Dutta, Industrial Robotics -Technology, Programming and Applications (SIE), 2nd Edition, McGraw Hill Book Co., Singapore, 2017
3	Ranky P.G. and Ho C.Y., Robots Modelling Control and Applications with Software, Springer Verlag, 1985
4	Craig J.J., Introduction to Robotics: Mechanics and Control, 3e, Pearson, 2008
5	Schilling R.J., Fundamentals of Robotics Analysis and Control, Prentice Hall of India, 1996
6	Saha S.K., Introduction to Robotics, McGraw-Hill Publication, 2008



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Course Name:	Non-Destructive Testing Methods		
Course Code:	OE-ME801C	Category:	Open Elective Courses
Semester:	Eighth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To study and understand the various Non-Destructive Evaluation and Testing methods, theory and their industrial applications.		
2	To know properties of materials without destroying the materials		
3	To understand and analyze of different types of defects to find out the defectless material for proper design		
4	To apply various non-destructive concepts to measure the dimension		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Fundamentals and introduction to destructive and non-destructive testing , scope and limitations of non-destructive testing (NDT), application of NDT methods, classification of NDT methods.	6
2	Surface defect analysis NDT methods: Visual/Optical Examination: Principal, Procedure, Instrument, Applications. Dye penetrant Testing (DPT)/ Liquid Penetrant Testing (LPT): Principle, procedure, characteristics of penetrant, types of penetrants, application and limitations. Magnetic Particle Testing (MPT): principle, magnetizing technique, procedure, equipment, application and limitations. Eddy Current Testing: Principle, instrument, techniques, sensitivity, application, limitation.	8
3	Internal defect analysis NDT methods: Ultrasonic methods: principle, generation of ultrasonic waves, equipment details, methods of wave propagation, various methods of ultrasonic testing, different ultrasonic scanning methods, advantages and limitation. Radiography: principle of radiography, types of radiography, equipments for x-ray radiography, advantages and applications. Thermography and Accoustic emission testing.	8
4	Comparison and selection of NDT Methods: Inspection of Rawmaterials, Inspection of Secondary Processing, In-service DamageInspection.	8
5	Common Application of NDT methods: a) Characterization of materials, b) Defect analysis, c) measurement of dimension. Case study. Codes Standards, Specifications andProcedures.	6
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Select an appropriate NDT technique as per requirement.
2	Explain and differentiate various defect types and select the appropriate NDT methods for better evaluation.
3	Explain and calibrate the instrument and inspect for in-service damage in the components.
4	Determine the flaws in the material by NDT and take measures to eliminate them.

Learning Resources:

1	Baldev Raj, T. Jayakumar, M. Thavasimuthu, Practical Non-Destructive Testing, Narosa Publishing House, 2014.
2	Ravi Prakash, Non-Destructive Testing Techniques, 1 st revised edition, New Age International Publishers, 2010.
3	J. Prasad, C. G. K. Nair, Non-Destructive Test and Evaluation of Materials, Tata McGraw-Hill Education, 2 nd edition, 2011.
4	Charles, J. Hellier, Handbook of Nondestructive evaluation, McGraw Hill, New York 2001.
5	Ed. Peter. J. Shull, Nondestructive Evaluation: Theory, Techniques, and Applications, Marcel Dekker, 2002.
6	Hull B., V. John, Non-Destructive Testing, Macmillan, 1988.



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Course Name:		Comprehensive Viva Voce		
Course Code:		PC-AUE881	Category:	Professional Core Courses
Semester:		Eighth	Credit:	1
L-T-P:		0-0-0	Pre-Requisites:	Nil
Full Marks:		100		
Examination Scheme:		Semester Examination: 100		
Course Objectives:				
1	The objective of comprehensive viva-voce is to assess the overall knowledge, a student acquired in the relevant field of engineering over 4 years of study in the programme. In doing so, the main objective is to prepare the students to face interview both in the academic and the industrial sector.			

Course Contents:	
Description of Topic	
The Comprehensive Viva-Voce will be conducted by a Committee consisting of Head of the Department and all Faculty members of the department. The Comprehensive Viva-Voce is intended to assess the student's understanding of the courses he/ she studied during the 4 years B.Tech programme.	
Course Outcomes:	
After completion of the course, students will be able to:	
1	Student will be able to prepare for the interview in a better way by brushing up different course papers so that overall knowledge on Automobile Engineering areas would be sharpened.



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Course Name:	Project-IV		
Course Code:	PW-AUE882	Category:	Project/ Internships/ Sessional
Semester:	Eighth	Credit:	6
L-T-P:	0-0-12	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		
Course Objectives:			
1	To develop the ability to identify, formulate and analyze engineering problems through literature survey, recent trends in industries and by applying the knowledge of science and engineering fundamentals.		
2	To train students in preparing project reports, to face reviews and viva voce examination		

Course Contents:	
Description of Topic	Contact Hrs.
It is intended to start the project work and carry out both design and fabrication of a mechanical/automobile device whose working can be demonstrated. The design and formulation of the problem and the fabrication and demonstration will be carried out. The students in a group of 4 to 6 works on a topic.	144
Total	144
Course Outcomes:	
After completion of the course, students will be able to:	
1	carry out some project works based on some design or fabrication or experimental problems in a group building up team spirit and would get sufficient exposure for the way to proceed to solve a practical or design problem